

Library Record Management

A real-world problem you discovered in your college - explained as an engineering problem, not yet as a final project.

Core observation: Students manually write their library entry and exit details in a physical register. The larger question is what happens to those records afterwards.

1. What is happening today?

Student enters	Writes in register	Register is stored
name + ID date + entry time	exit time + signature	paper records

The possible hidden workflow

The paper register may not be the end of the process. Staff may later need to read, search, count, verify, or manually transfer information into a spreadsheet or computer system. That second stage can create a larger operational burden than the writing itself.

Physical register	Manual checking	Possible re-entry	Spreadsheet / records
handwritten data	find old entries	type data again	search / reports

Key insight: The project may be less about replacing a pen-and-paper register and more about removing an inefficient paper -> human -> computer data pipeline.

Important: this is still a hypothesis. Before building anything, verify how your library actually handles the records after students write them.

2. What problems could exist?

Problem area	What it could look like	Why it matters
Data entry	Someone reads handwritten records and types them into Excel/PC.	Duplicate work and human effort.
Searching old records	Staff must open old registers and scan pages for a student/date.	Slow and difficult as records grow.
Data quality	Unreadable handwriting, wrong ID, missing exit time, inconsistent formatting.	Records become harder to trust or use.
Delayed digital data	Computer records are created only after manual entry.	Information is not immediately available.
Reporting	Counts, busiest hours, visit duration, or summaries require manual work.	Useful information takes extra effort to produce.
Physical storage	Registers accumulate over months/years.	Space, preservation, and retrieval become harder.

3. The problem statement is not fixed yet

You should not start by deciding that the answer is RFID, QR, face recognition, OCR, or an app. Those are technologies. First establish which part of the workflow is actually causing the most pain.

Question to investigate	What you are trying to discover
Do staff manually transfer register data to a computer?	Whether a second manual data-entry step really exists.
How are old records searched?	The real retrieval burden and how often it happens.
How are missing/incorrect entries handled?	Where errors enter the workflow.
What reports are needed?	Whether there is a reporting/analytics problem.
Who spends time on this process?	The actual user who experiences the problem.
How often does the problem occur?	Whether it is occasional or continuous.

4. Two possible project directions

Once you validate the workflow, two broad directions make sense. They solve different problems.

A. Prevent the problem	B. Fix the existing backlog
Digital entry system Capture a student ID and timestamp digitally so a paper register and later manual re-entry are no longer necessary.	Record digitization system Take existing paper records and convert them into structured digital data using OCR, extraction, validation, and export.
Example flow Student ID -> timestamp -> database -> search/dashboard	Example flow Register scan/photo -> OCR -> extracted fields -> validation -> database/Excel

Which one is more interesting?

Do not decide by how “cool” the technology looks. Decide by the real workflow. If staff repeatedly type handwritten records into a PC, digitization may directly address their current workload. If the main issue is creating and retrieving records in the future, a digital entry system may address the root cause.

Best engineering practice: Build the smallest prototype that proves the main idea. Do not build a full college deployment before proving that the workflow actually needs it.

5. What a strong final project could demonstrate

Layer	Possible demonstration
Input	QR, RFID, barcode, or another identification method
Processing	Validation, duplicate detection, entry/exit state logic
Storage	Database with searchable visit records
Interface	Simple student/staff dashboard
Automation	Automatic timestamps and record creation
Analytics	Daily visitors, active occupancy, visit duration, peak usage
Reliability	Handling duplicate scans, missing exits, connectivity/power issues

6. Your field investigation checklist

Before coding, spend a little time understanding the real process. Your goal is to discover the highest-friction step, not to prove your first idea is correct.

Observe	Watch the complete process from student entry to record storage/use.
Ask	Talk to staff who actually maintain or retrieve the records.
Measure	Estimate entries per hour/day, time spent per entry, and how often old records are requested.
Trace	Follow one record through the system: where it starts, who touches it, and where it ends.
Find pain	Look for repetition, searching, retyping, missing data, delays, and manual reporting.
Define	Write one specific problem statement based on evidence you observed.

A useful problem-statement template

"In our library, [person] currently spends [time/effort] doing [manual task] because [current process]. This creates [specific consequence]. I want to investigate whether [proposed improvement] can reduce that burden while keeping the records accurate and searchable."

Example - only after verification

"Library staff currently maintain student visit records through handwritten registers and later manual data entry. Retrieving historical visits and converting paper entries into digital records takes repeated human effort and can introduce errors. A prototype could automate record capture or digitization and provide searchable records."

Your immediate next step

Do not build yet.

First find out exactly what happens to the library register after the page is filled. That answer will determine whether your strongest project is a digital entry system, an OCR-based digitization tool, or something else.

This document is a problem-discovery brief, not a claim that your library currently uses any particular workflow. The possible PC/data-entry step should be verified on-site before it becomes part of the final problem statement.